



Student Number	
Mark / 35	

Chemistry

Production of Materials

Theory Test • 2003

General Instructions

- Reading time – 5 minutes
- Working time – 55 minutes
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A Data Sheet and a Periodic Table are provided at the back of this paper
- Write your Student Number at the top of this page

Total Marks – 35

Part A – 10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this part

Part B – 25 marks

- Attempt Questions 11 – 15
- Allow about 40 minutes for this part

Allow about 15 minutes for this part

A  B  C  D 

► Mark your answers for Questions 1 – 10 in the Answer Box on page 1.

- 1 Which of the following is a transuranic element?
- (A) bohrium
 - (B) thallium
 - (C) thorium
 - (D) thulium
- 2 Which of the following occurs when a polymer is formed by condensation polymerisation?
- (A) The mass of the polymer formed is less than the combined mass of the reactants.
 - (B) It becomes a mixed polymer.
 - (C) Only one product is formed in the reaction.
 - (D) One product must always be water.
- 3 Ethanol is a solvent for many substances. Which of the following statements is an **incorrect** explanation of ethanol's solubility?
- (A) Ethanol has an OH group which helps it dissolve polar molecules.
 - (B) Ethanol can form hydrogen bonds with water.
 - (C) Ethanol has a CH₃CH₂ chain which helps it form covalent bonds with non-polar substances.
 - (D) Ethanol has an OH group which helps it dissolve ionic substances.
- 4 Which of the following is a monomer for cellulose?
- (A) β-glucose
 - (B) β-cellulase
 - (C) starch
 - (D) sucrose
- 5 A student correctly sets up an experiment to convert glucose into ethanol. She monitored the mass of the reaction flask over a few days and found that her reaction flask decreased in mass by 4.4 grams. What mass of ethanol was produced?
- (A) 0 g
 - (B) 4.4 g
 - (C) 4.6 g
 - (D) 9.2 g

6 How can ethylene be obtained from crude oil?

- (A) By separating out the lighter components by fractional distillation.
- (B) By separating out the heavier components by fractional distillation.
- (C) By catalytic cracking of the crude oil followed by distillation.
- (D) By decomposing the crude oil followed by distillation.

7 A student burns ethanol in a spirit burner to heat 150 mL of water. His results are...

Initial temperature of water	24.5 °C
Final temperature of water	74.5 °C
Initial mass of burner + ethanol	236.3 g
Final mass of burner + ethanol	234.3 g

What is the heat of combustion per gram of ethanol from this student's results?

- (A) 31,350 kJ
- (B) 15,675 J
- (C) 418 J
- (D) 31,350 J

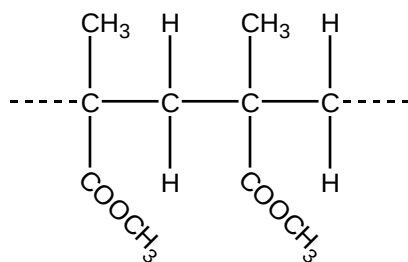
8 Which list shows the metals in order of increasing activity according to the standard potentials data?

- (A) Ag, Fe, Cu, Ni
- (B) Fe, Al, Mn, Ca
- (C) Pb, Fe, Ca, Na
- (D) Cu, Mn, Na, Ba

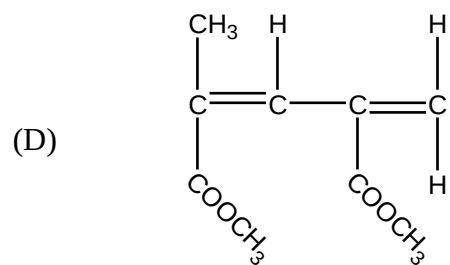
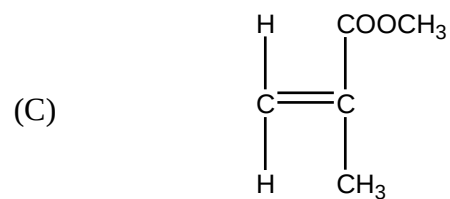
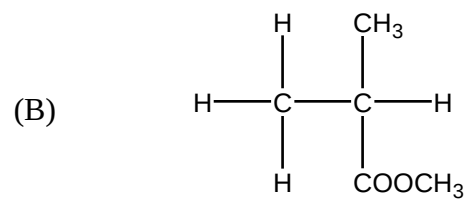
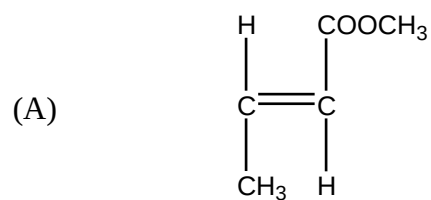
9 In which species is manganese in the lowest oxidation state?

- (A) MnO_4^{2-}
- (B) MnO_4^-
- (C) MnO
- (D) Mn_2O_3

10 An addition polymer used in the manufacture of artificial eyes has the structure...



Which of the following chemicals is the monomer for this polymer?



Part B – 25 marks

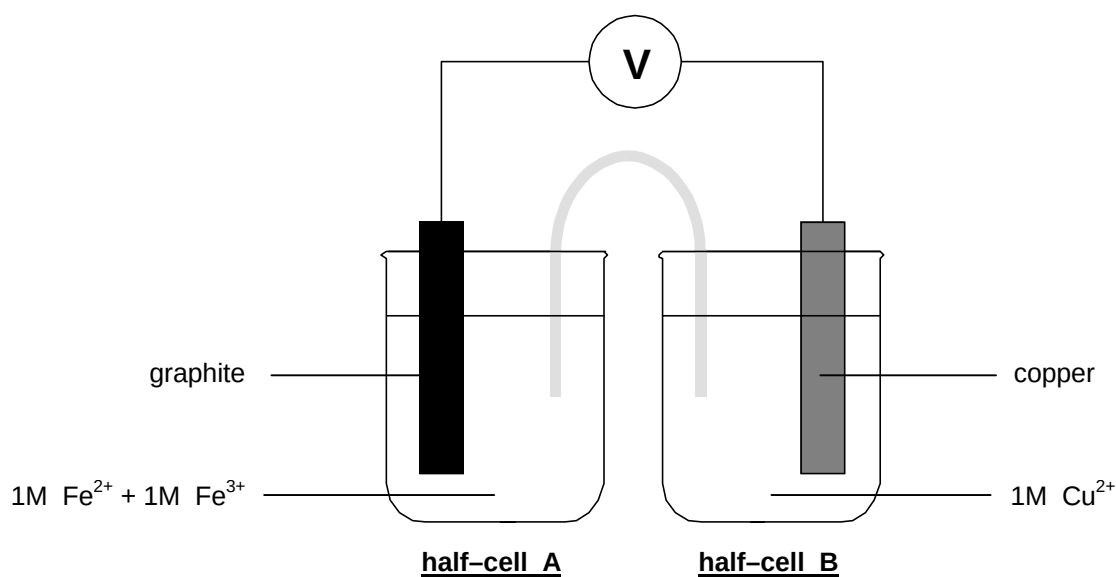
Attempt Questions 11 – 15

Allow about 40 minutes for this part

► **Show all relevant working in questions involving calculations.**

Question 11 (4 marks)

The diagram shows a galvanic cell composed of two half-cells connected by a salt bridge...



After operating for several hours chemical changes are evident.

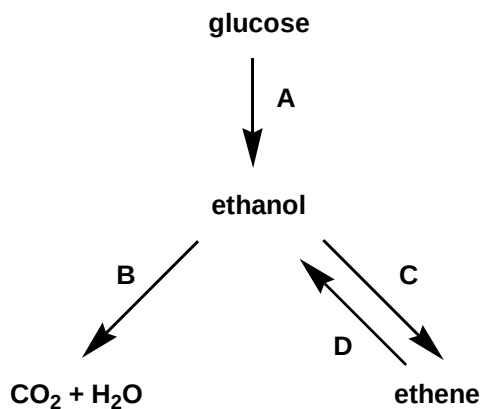
The reaction occurring in half-cell A is... $\text{Fe}^{3+}_{(\text{aq})} + \text{e}^{-} \rightarrow \text{Fe}^{2+}_{(\text{aq})}$

- (a) Indicate the direction of electron flow on the diagram. **(1 mark)**
- (b) Describe two changes that would be visible in half-cell B after several hours? **(2 marks)**

- (c) Calculate the net voltage of the galvanic cell. **(1 mark)**

Question 12 (9 marks)

Identify the type of reaction (A, B, C & D) in the flow chart and write a balanced chemical equation for each reaction. ► *Include states of matter and conditions.*



Reaction	Type of Reaction (4 marks)
A	
B	
C	
D	

Reaction	Chemical Equation (5 marks)
A	
B	
C	
D	

Question 13 (4 marks)

- (a) Describe two conditions under which a nucleus is unstable. **(2 marks)**

- (b) What is the effect of a nucleus being unstable? **(1 mark)**

- (c) Identify an instrument that could be used to detect a substance that has unstable nuclei. **(1 mark)**

Question 14 (5 marks)

- (a) Give an equation (using structural formulae) for the reaction between ethylene and bromine water and name the organic product. **(2 marks)**

- (b) (i) Identify the systematic name for styrene. **(1 mark)**

- (ii) Describe **one** use for polystyrene and identify a property which makes it useful for this purpose. **(2 marks)**

Question 15 (3 marks)

Complete the table for either a dry cell or lead–acid cell...

	TYPE OF CELL Dry cell <u>or</u> Lead–acid cell (circle your choice above)
Identify the composition of the anode	
Write the reduction half–equation	
<u>One</u> advantage of the cell	

HIGHER SCHOOL CERTIFICATE EXAMINATION
Chemistry

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+] \qquad \Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

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1 H 1.008 Hydrogen																	He 4.003 Helium	
		KEY																
		Atomic Number																
		Atomic Weight																
		Symbol of element																
		Name of element																

Lanthanides

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [146.9] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinides

89 Ac [227.0] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237.0] Neptunium	94 Pu [239.1] Plutonium	95 Am [241.1] Americium	96 Cm [244.1] Curium	97 Bk [249.1] Berkelium	98 Cf [252.1] Californium	99 Es [252.1] Einsteinium	100 Fm [257.1] Fermium	101 Md [258.1] Mendelevium	102 No [259.1] Nobelium	103 Lr [262.1] Lawrencium
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Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.



ANSWERS and MARKING SCHEME

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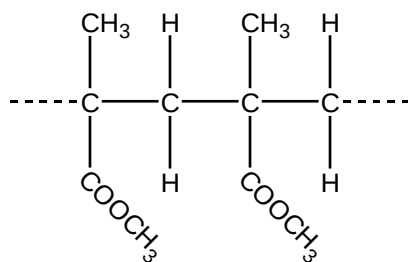
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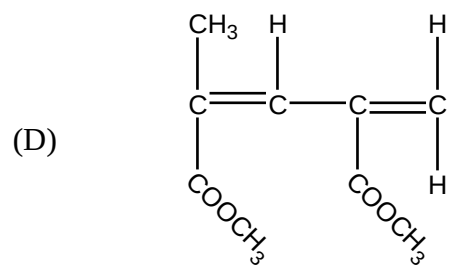
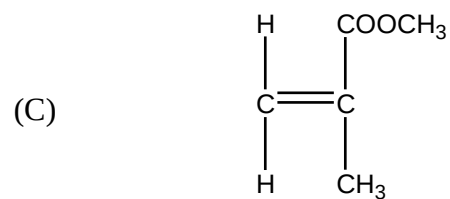
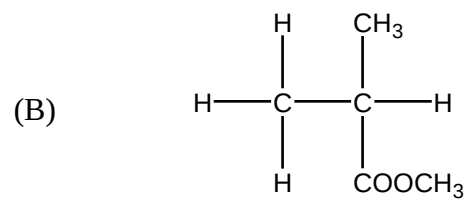
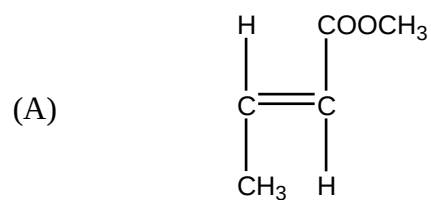
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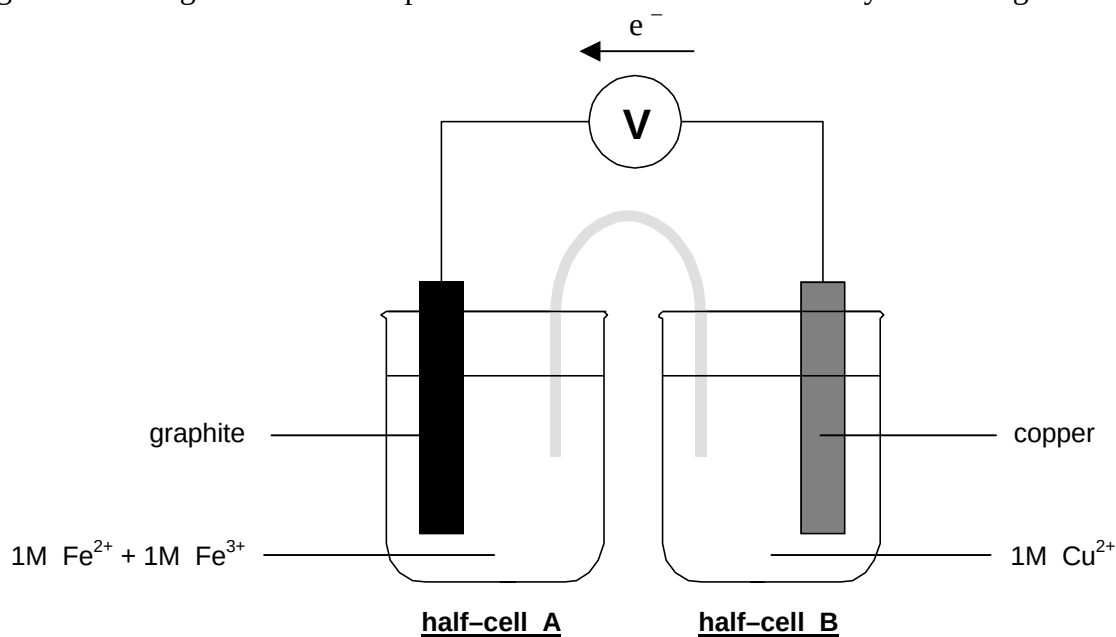
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Question 11 (4 marks)

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After operating for several hours chemical changes are evident.

The reaction occurring in half-cell A is... $\text{Fe}^{3+}_{(\text{aq})} + \text{e}^{-} \rightarrow \text{Fe}^{2+}_{(\text{aq})}$

- (a) Indicate the direction of electron flow on the diagram. **(1 mark)**
- (b) What changes would be visible in half-cell B after several hours? **(2 marks)**

The copper electrode becomes thinner. (1 mark)

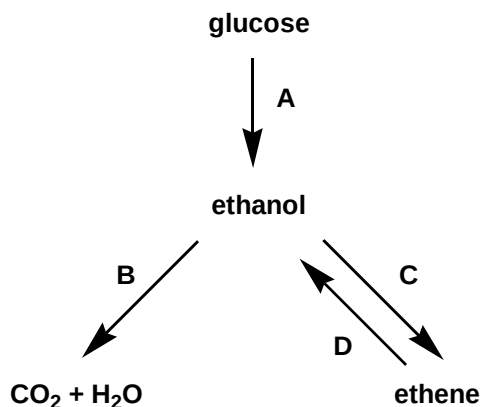
The blue colour of the Cu²⁺ electrolyte becomes darker. (1 mark)

- (c) Calculate the net voltage of the galvanic cell. **(1 mark)**

$$E^{\circ}_{\text{net}} = E^{\circ}_{\text{ox}} + E^{\circ}_{\text{red}} = -0.34 \text{ V} + 0.77 \text{ V} = +0.43 \text{ V}$$

Question 12 (9 marks)

Identify the type of reaction (A, B, C & D) in the flow chart and write a balanced chemical equation for each reaction. ► *Include states of matter and conditions.*



Reaction	Type of Reaction (4 marks)
A	Fermentation
B	Combustion
C	Dehydration
D	Addition or hydration

Reaction	Chemical Equation (5 marks)
A	$\text{C}_6\text{H}_{12}\text{O}_6 (\text{aq}) \xrightarrow{\text{Yeast (zymase)}} 2\text{CO}_2 (\text{g}) + 2\text{C}_2\text{H}_5\text{OH} (\text{aq})$
B	$\text{C}_2\text{H}_5\text{OH} (\text{l}) + 3\text{O}_2 (\text{g}) \rightarrow 2\text{CO}_2 (\text{g}) + 3\text{H}_2\text{O} (\text{l or g})$
C	$\text{C}_2\text{H}_5\text{OH} (\text{l}) \xrightarrow{\text{conc. H}_2\text{SO}_4 + \text{heat}} \text{C}_2\text{H}_4 (\text{g}) + \text{H}_2\text{O} (\text{l or g})$
D	$\text{C}_2\text{H}_4 (\text{g}) + \text{H}_2\text{O} (\text{g}) \xrightarrow{\text{H}_2\text{SO}_4 \text{ or } \text{H}_3\text{PO}_4 + \text{heat}} \text{C}_2\text{H}_5\text{OH} (\text{l})$

Question 13 (4 marks)

(a) Describe two conditions under which a nucleus is unstable. **(2 marks)**

(i) **The ratio of neutrons to protons is unstable. (1 mark)**

(ii) **Large atomic number (greater than 83). (1 mark)**

(b) What is the effect of a nucleus being unstable? **(1 mark)**

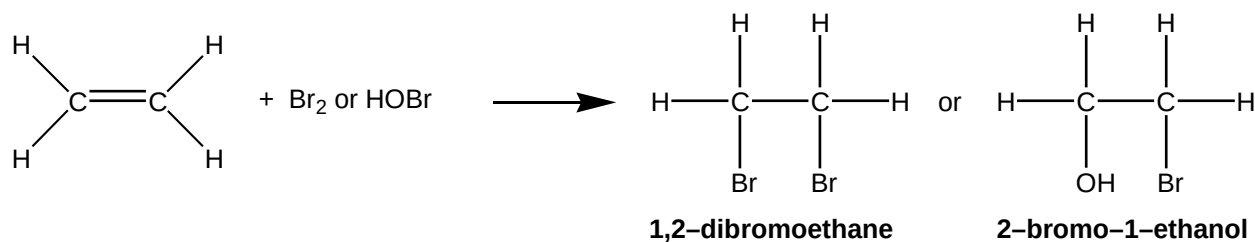
Emission of radioactivity or α or β or γ emission or undergoes radioactive decay.

(c) Identify an instrument that could be used to detect a substance that has unstable nuclei. **(1 mark)**

Geiger counter or Cloud chamber or Bubble chamber

Question 14 (5 marks)

- (a) Give an equation (using structural formulae) for the reaction between ethylene and bromine water and name the organic product. **(2 marks)**



- (b) (i) Identify the systematic name for styrene. **(1 mark)**

ethenylbenzene

phenylethene

- (ii) Describe one use for polystyrene and identify a property which makes it useful for this purpose. **(2 marks)**

Polystyrene can be used for....

- television backings because of its electrical insulating properties.**
- disposable cups for hot/cold drinks because of the excellent thermal insulation of PS foam.**
- lens (magnifying glasses and cheap cameras) because it is transparent and easily moulded.**
- CD cases and model aircraft, etc. because it can be precisely and easily moulded by injection.**

(1 mark for use. 1 marks for property.)

Question 15 (3 marks)

Complete the table for either a dry cell or lead–acid cell...

	TYPE OF CELL Dry cell <u>or</u> Lead–acid cell (circle your choice above)
Identify the composition of the anode	Dry cell's anode is zinc. Lead–acid cell's anode is lead.
Write the reduction half–equation	Dry cell – $\text{NH}_4^+ + \text{MnO}_2 + \text{H}_2\text{O} + \text{e}^- \rightarrow \text{Mn(OH)}_3 + \text{NH}_3$ Dry cell – $2\text{NH}_4^+ + 2\text{MnO}_2 + 2\text{e}^- \rightarrow \text{Mn}_2\text{O}_3 + 2\text{NH}_3 + \text{H}_2\text{O}$ Lead–acid – $\text{PbO}_2 + \text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \rightarrow \text{PbSO}_4 + 2\text{H}_2\text{O}$
<u>One</u> advantage of the cell	Dry cell is inexpensive; can be safely disposed when 'dead'. Lead–acid cell provides high current output; reliable; relatively inexpensive; rechargeable.